

HAZEM MARRAKCHI

Software Engineer | Full Stack Developer

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GitHub : <https://github.com/HazemMarrakchi>

Portfolio : <https://hazemmarrakchi.github.io/portfolio/> | LinkedIn : <https://www.linkedin.com/in/hazem-marrakchi/>

PROFILE

Software engineering graduate from ESSAT Gabès (EUR-ACE® accredited programme), I design and build modern full-stack web applications, from architecture modelling to end-to-end testing. I primarily work within the TypeScript ecosystem — React on the frontend, Node.js and NestJS on the backend — with solid hands-on experience in Java Spring Boot and Python FastAPI gained through industry internships. What sets me apart: I turned a complex European regulation, EN 16931 e-invoicing, into two real products deployed online — Facturia and Invoice Studio. Autonomous and rigorous, I treat quality as part of development itself: CI/CD with GitHub Actions, automated Playwright and Vitest tests, Docker containerization.

PROJECTS

FACTURIA — EU-Compliant E-Invoicing SaaS Platform

Personal project

Live demo : <https://hazemmarrakchi.github.io/facturia/> | Source : <https://github.com/HazemMarrakchi/facturia>

Stack: React 19 • TypeScript • NestJS • Drizzle ORM • SQLite • Zod • Tailwind CSS v4 • Docker

- Generation of UBL 2.1 e-invoices compliant with Peppol BIS Billing 3.0 and XRechnung, automatically validated against EN 16931 business rules.
- Complete invoice lifecycle (draft, sent, accepted, refused, paid) with per-organization sequential numbering and client directory.
- Multi-organization architecture with role-based access control and an immutable GDPR audit trail including data export and deletion.
- REST API built with NestJS and documented with OpenAPI/Swagger; React frontend deployed on GitHub Pages consuming a real API hosted on Render.
- Quality: 8 Playwright End-to-End tests running against the live server; CI/CD pipeline with GitHub Actions; Docker images.

EINVOICE STUDIO — Public EN 16931 E-Invoice Validator

Personal project

Tool : <https://hazemmarrakchi.github.io/einvoice-studio/> | Source : <https://github.com/HazemMarrakchi/einvoice-studio>

Stack: React 19 • TypeScript • Vite • Tailwind CSS v4 • Vitest

- Fully client-side web application: instant rule-by-rule diagnostics (BR-*, BR-CO-*) with clear explanations and concrete fix hints.
- Visual structure explorer: parties, totals, VAT breakdown and invoice lines parsed into readable cards.
- Pure-TypeScript rule engine with a namespace-agnostic XML parser; complete privacy (no data ever leaves the machine).
- Vitest unit tests gating deployment in CI.

DHAHABI — Online Loan Application Platform

ZeTheta Program — Fintech Project

Demo : <https://hazemmarrakchi.github.io/loan-application/> | Source : <https://github.com/HazemMarrakchi/loan-application>

Stack: React 19 • TypeScript • Vite • Tailwind CSS v4 • React Hook Form • Zod • Playwright

- A 9-step loan application wizard for the Tunisian market: real-time validation, conditional fields per loan type and client-side document compression.
- E-signature capture, auto-save with resume capability and an instant pre-approval engine (40% repayment rule).
- Quality: fully covered by 8 Playwright End-to-End tests.

INTELLIGENT HOSPITAL MANAGEMENT PLATFORM — Final Year Project (Digilife)

Academic project

Stack: Angular • Node.js • Express.js • FastAPI • MongoDB • WebSockets • JWT

- Real-time coordination between hospital services via WebSockets together with AI-based prediction features.
- Grade: 18.5/20 — High Honors.

CINEMA TICKET BOOKING PLATFORM — Engineering Degree Project

Academic project

Stack: Angular • Spring Boot • MySQL

- Session management, seat selection and a REST Spring Boot backend backed by MySQL.

CARPOOLING WEB APPLICATION — Bachelor's Final Year Project

Bachelor's in Business Computing

Stack: PHP • JavaScript • Bootstrap • MySQL

- Trip publishing, route search and booking management connecting drivers and travellers.

PROFESSIONAL EXPERIENCE

Software Engineering Intern Feb 2026 – Jun 2026

Digilife, Gabès

- Designed and developed an intelligent hospital management platform integrating AI-based predictions (Angular, Node.js, Express.js, FastAPI, MongoDB).
- Contributed to software architecture design and to the development of JWT-secured REST APIs.

Software Engineering Intern Jul 2025

Telecom Operator, Gabès

- Developed web and mobile applications, showcase websites and internal tools.
- Deployed and maintained internal IT and application infrastructure.

Software Engineering Intern Jul 2024

Telecom Operator, Gabès

- User support and maintenance of IT assets, network and messaging systems.
- Built database queries and kept data up to date; proposed automation solutions.

IT Intern Jul 2022

STEG, Gabès

- IT maintenance and support activities; assisted with IT environment management.

Freelance Web Developer Apr 2018 – Oct 2020

Sana Travel Agency, Gabès

- Built and maintained the showcase website and the online booking platform (front-end and back-end).
- Integrated airline, hotel and payment provider APIs.
- SEO and performance optimization; security and site updates.

EDUCATION

- 2023 – 2026: National Degree in Software Engineering — ESSAT Gabès, Tunisia (EUR-ACE® accredited programme, ASIIN e.V.)
- 2020 – 2023: Bachelor's Degree in Business Computing — ESSAT Gabès, Tunisia
- 2017 – 2018: Baccalaureate in Economics and Management — Gabès, Tunisia

TECHNICAL SKILLS

- Languages: TypeScript, JavaScript, Java, Python, SQL, PHP
- Frontend: React, Angular, Tailwind CSS, HTML5/CSS3, Bootstrap
- Backend: NestJS, Node.js, Express.js, Spring Boot, FastAPI, REST APIs, WebSockets
- Databases: MySQL, MongoDB, SQLite, Oracle Database, Drizzle ORM
- DevOps & Cloud: Docker, GitHub Actions (CI/CD), Linux, Microsoft Azure, Render
- Testing & Quality: Playwright E2E, Vitest, OpenAPI/Swagger, Zod
- AI & Machine Learning: Scikit-learn, Deep Learning
- Tools & Methods: Git, GitHub, Postman, Agile Scrum, UML, n8n

LANGUAGES AND CERTIFICATIONS

- Languages: French (professional), English (professional), Arabic (native), German (beginner A1)
- Certifications: Full Stack and Front End Development (LinkedIn Learning), Deep Learning Foundation